高等数学 IB 第一次作业

（满分：100 分）

作业提交截止时间：**2023 年 5 月 22 日**

作业提交方式：线下提交到 25 教 1808 或在课堂上提交，**不接受电子版**

（重修同学手写编辑为 PDF 文档发送到 QQ 邮箱 503263897@qq.com）。

答题要求：1. **必须写明解题步骤，否则只有结果分**.

2. **答题时用作业纸手写**.

1. (10 marks). Suppose that the equation $2x^2+2y^2+z^2+8yz-z+8=0$ defines the function $z=z(x, y)$ and satisfies $z(0, -2)=1$. Find $\frac{\partial z}{\partial x}(0,-2)$ and $\frac{\partial^2 z}{\partial x^2}(0,-2)$.

2. (10 marks) If $a_n=\frac{(-1)^n}{\sqrt{n}}$, which of the following series is absolutely convergent? (　　) Given reasons for your answer.

(A) $\sum_{n=1}^{\infty}(-1)^n a_n$　　(B) $\sum_{n=1}^{\infty} a_n a_{n+1}$　　(C) $\sum_{n=1}^{\infty}(a_{n+1}-a_n)$

(D) $\sum_{n=1}^{\infty}(a_{n+1}+a_n)$

3. (15 marks) Find the sum of the series $\sum_{n=1}^{\infty}\frac{n^2}{2^{n-1}}$.

4. (10 marks) Find the derivative of the function $u=\frac{x^2\ln y}{z}$ at the point M (1, 1, 2) in the direction of $\boldsymbol{n}$, where $\boldsymbol{n}$ is the normal vector of the surface $z=x^2+y^2$ at M pointing downwards.

5. (20 marks) Given the surface S: $2x^2+4y^2+z^2=4$ and the plane II: $2x+2y+z+5=0$. Find: (1) the tangent plane of S which is parallel to II; (2) the shortest distance between S and II.

6. (10 marks) Given three lines: $x=0$, $y=0$, and $x-y+1=0$. Find a point M (x, y) on the xy-plane, such that the sum of the squared distances between M and the three lines is smallest.

7. (10 marks) If $f(x, y)=x^4-3x^2y+2y^2$, which of the following statements is true? (　　) Given reasons for your answer.

(1) $x=0$ is the point where $f(x,0)$ has a local minimum value; (2) $y=0$ is the point where $f(0,y)$ has a local minimum value; (3) $x=0$ is the point where $f(x, kx)$ ($k\neq 0$) has a local minimum value; (4) (0, 0) is the point where $f(x, y)$ has a local minimum value.

(A) Only (1) and (2) are correct　(B) only (3) is correct　(C) only (1), (2), and (3) are correct　(D) (1)-(4) are all correct.

8. (15 marks) Let $u=f(x, y, z, t)$ be a differentiable function, where $z=z(y)$ and $t=t(y)$ are determined by the equations $\begin{cases} y^2+yz-zt^2=0 \\ t+2z=0 \end{cases}$, find $\frac{\partial u}{\partial y}$ and $\frac{\partial^2 u}{\partial x\,\partial y}$.